

Building Classification From Aerial Imagery

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Abstract—This document is a template demonstrating the Ieee-conference format.

Index Terms—template; demo

I. INTRODUCTION

Travel forecasting models are based heavily on data provided by government statistical agencies, such as the US Census Bureau and the Bureau of Labor Statistics. These agencies provide the land use data needed to construct these models, giving statistics such as household number, median household income, etc. for census areas. In countries where statistical agencies are not as reliable, travel forecasting models are generally not developed or have data gaps. In countries where they are available, these models are highly reliant on the public release of such statistical data, which leads to problems if such surveys are delayed or canceled.

Recent research has sought to find other ways to collect land use characteristics

Talk about research using satellites and other publicly available data sets. What data is currently being collected and analyzed?

Remote Sensing Data (From Satellites) [1] - Night time Light data is strongly correlated with social economic variables such as population, GDP and electric power consumption. [2] - Night time light from vehicles to predict car travel between cities on highways

Economic activity [3] - Developing proxy economic activity from daytime imagery for periods without the data. And AI model was developed to assist in the analysis.

Estimate populations [4] - Uses RGB images (From satellites?), road length and density, and distance from populations to estimate populations in rural areas. Density of Roads is correlated to population. [5] - Used deep learning to estimate population characteristics and location of human made structures. Area, volume, and inhabitant number was estimated from AI model trained on GeoDenmark, lidar data, and population data.

Talk about drone use in collecting remote sensing data. With the rise of commercially available and affordable aerial drones,

the ability to capture remote sensing data such as high quality aerial imagery has become significantly easier.

Developing opensource software for drone use [6] - Opensource hardware and software for drone use.

Building Footprints [7] -Using AI model YOLO, object detection deep learning was used to extract building footprints. [8] - Extracting building footprints from point cloud data. Building were classified before extracting the footprints. [9] - Inventoried residential structures based off microsoft building footprint data, LiDAR, and point of interest data. When from census level data to anonymized individual house holds.

Land use [10] -remote sensing data to classify land use in China. Estimated landscape patterns and building functions, divided vegetation, water, and hard surfaces, categorized the buildings.

This leads to the question “can remote sensing data be used to develop land use data to supplement or replace data from statistical agencies?” In this paper the process, and the results of the trained model will be compared to other cities in Utah.

This paper presents a model that recognizes and categorizes building types based off remote sensing data and estimates household statistics for a general area.

II. METHODOLOGY

A. Data Collection

The city of Logan, Utah was chosen for analysis. Building footprints for Logan were obtained from the Microsoft Building Footprint public dataset. This dataset allowed for easy access to building footprint data for this initial study.

The latitude and longitude of the city boundaries was taken. All building footprints in the city boundary were extracted from the Microsoft Building Footprint dataset and saved as a geojson file. K-mean analysis was run to separate different sizes of buildings. Optimal K value was found using the elbow method. For Logan the number is five. K-mean analysis was run comparing the building footprint areas to their perimeters. As seen in Figure 1...

Using the building footprint location and “contextily”, 600 randomly chosen aerial images were taken from the internet and cropped to focus on a single building. 200 of the images

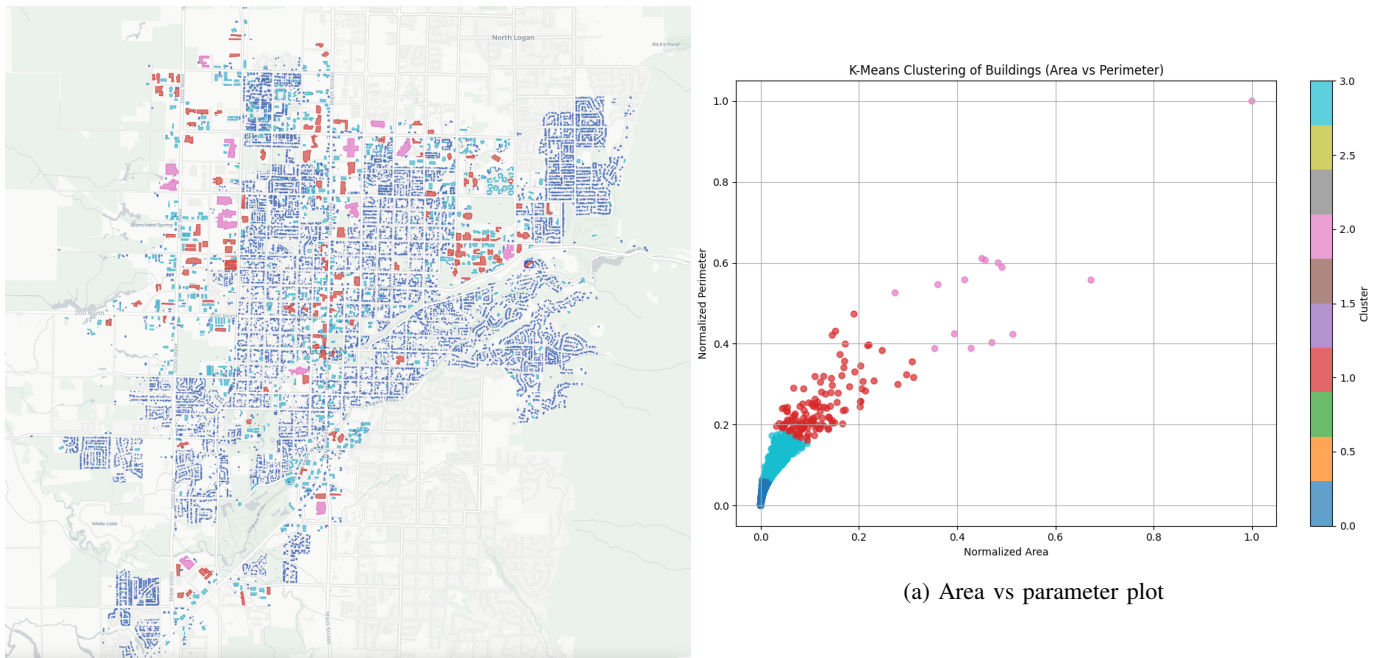


Figure 1: K-mean analysis for Logan, Utah

were from clusters that included only larger buildings to help balance building types in the dataset.

Using open source tool Label Studio, the images were manually annotated. Every building in each picture was labeled into five building categories: Apartments, Residential, Commercial, Church, and School.

B. Model Training

Training the Ultralytics You Only Look Once (YOLO) model based off labeled images

Images were split, 80% used in training and 20% in validation.

Image sets with clusters that included larger building sizes were specifically chosen to make sure the model would be trained on these less common building types.

C. Model Validation

Using trained model to predict building types in Logan Utah

Choosing cities in Utah using the American Community Survey data. Based off median income level and age of homes.

Using the model to predict building types in other cities in Utah based.

III. RESULTS AND DISCUSSION

Confusion matrix for Logan - Training and Validation As seen in Figure 2...

The model could identify accurately most building in Logan. It struggled with schools and churches. This is likely due to the very low number of images of both building types provided during training.

Confusion matrix for other analysis cities. What are the results? Was there a difference between how the model did

between the cities? Was age of the house a big factor? Was income? Is there something else?

Will it be easier to develop a model to be used at different cities, or a process for people to develop their own models?

Talk about how this data is still subject to many of the problems the survey data still is. However, much of this research is susceptible to similar problems as transportation models. Satellite remote sensing data is not consistent around the world,

1) *Subsubsection Heading Here*: Subsubsection text here.

IV. CONCLUSION AND FUTURE RESEARCH

The conclusion goes here.

As discussed by Hall et al., there are potential ethical problems with such data collection. It straddles the line between public and private information. This will need to be considered as further research is done into this area as to stay on the right side of the law.[11]

What does the future research look like?

Training models to detect other things such as cars, swing sets, patios, etc.

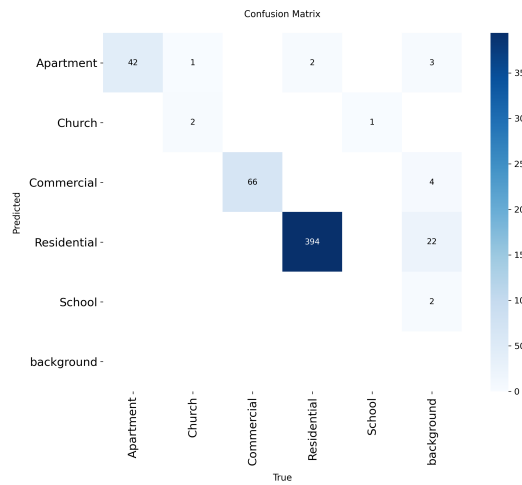
Creating the process to predict household types based off the collected data

Incorporating drone aerial imagery and extracting building footprints from that.

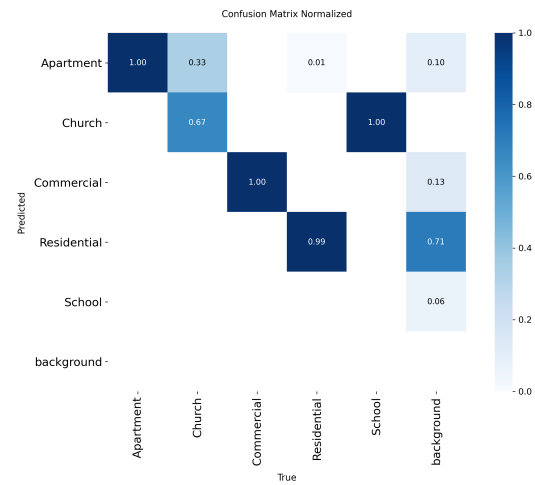
Research into other remote sensing data and how it could be incorporated.

V. ACKNOWLEDGMENT

The authors would like to thank...



(a) Logan confusion matrix



(b) Logan confusion Matrix normalized

Figure 2: YOLO confusion matrixes for Logan, Utah.

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